NYC Citi Bike gap analysis: table flow

One physical BigQuery table is reshaped into station metrics and a peer benchmark.

citibike_trips

SOURCE GRAIN: one recorded trip

- starttime, stoptime
- tripduration
- start_station_id / name
- end_station_id / name
- bikeid for duplicate checks

1 →

station_events

DERIVED GRAIN: one trip endpoint

- station_id
- station_name
- is_departure
- Each trip becomes 2 events
- Null endpoint IDs excluded

2 →

station_metrics

DERIVED GRAIN: one station

- station_id (grouping key)
- canonical station_name
- departures, arrivals
- total_station_events
- net_departures
- net_imbalance_rate

3 →

gap_benchmark

FINAL GRAIN: one target comparison

- target_station_id
- peer_station_count
- peer median abs. rate
- benchmark_abs_events
- excess_abs_events

Relationship note

The CP1 query does not join separate physical tables. 1: expand each trip into start and end events. 2: group events by station_id. 3: compare the target with the peer median. Station ID connects endpoint events to the station aggregate.